

A NOVEL MULTI-FOCUS FUSION NETWORK FOR RETINAL MICROSURGERY



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Introduction

Retinal Microsurgery

Requires high precision



Exists partially out-of-focus regions

High magnification

Shallow Depth-of-Field

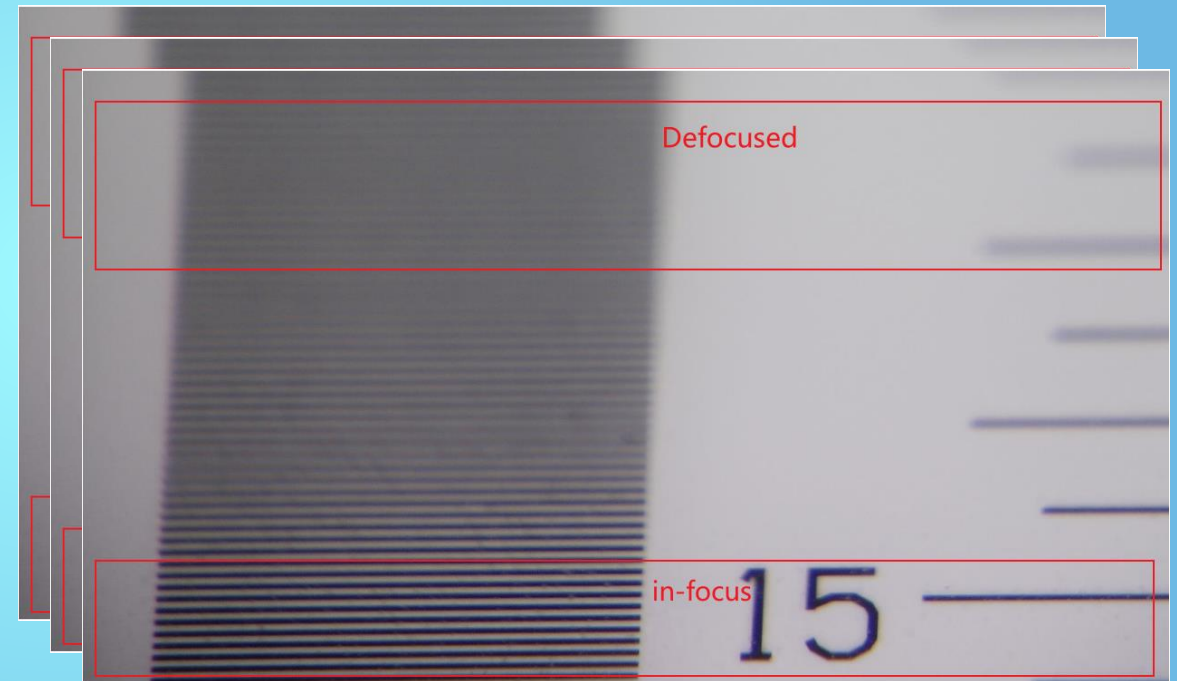
Eye's curvature

Multi-focus Fusion

A stack of defocused images
captured using different focal
settings



Reconstruct
All-in-focus Image



Related Work

Transform domain-based

Image Transform -> Coefficient Fusion -> Inverse Transform

For example: Gradient domain transform

Spatial domain-based

Weight Map Generation -> Weighted Source Images Fusion

For example: Block-based, Region-based, Pixel-based

Deep learning-based

Classification: Focus Map Generation -> Decision Map Generation -> Image Fusion;

Regression: Feature Extraction -> Feature Fusion -> Feature Reconstruction

For example: U-Net based

Problems

Time consuming

Difficult in figuring out the boundary between out-of-focus ones and in-focus ones

Low performance in representing contextual features for medical images

Our Solution

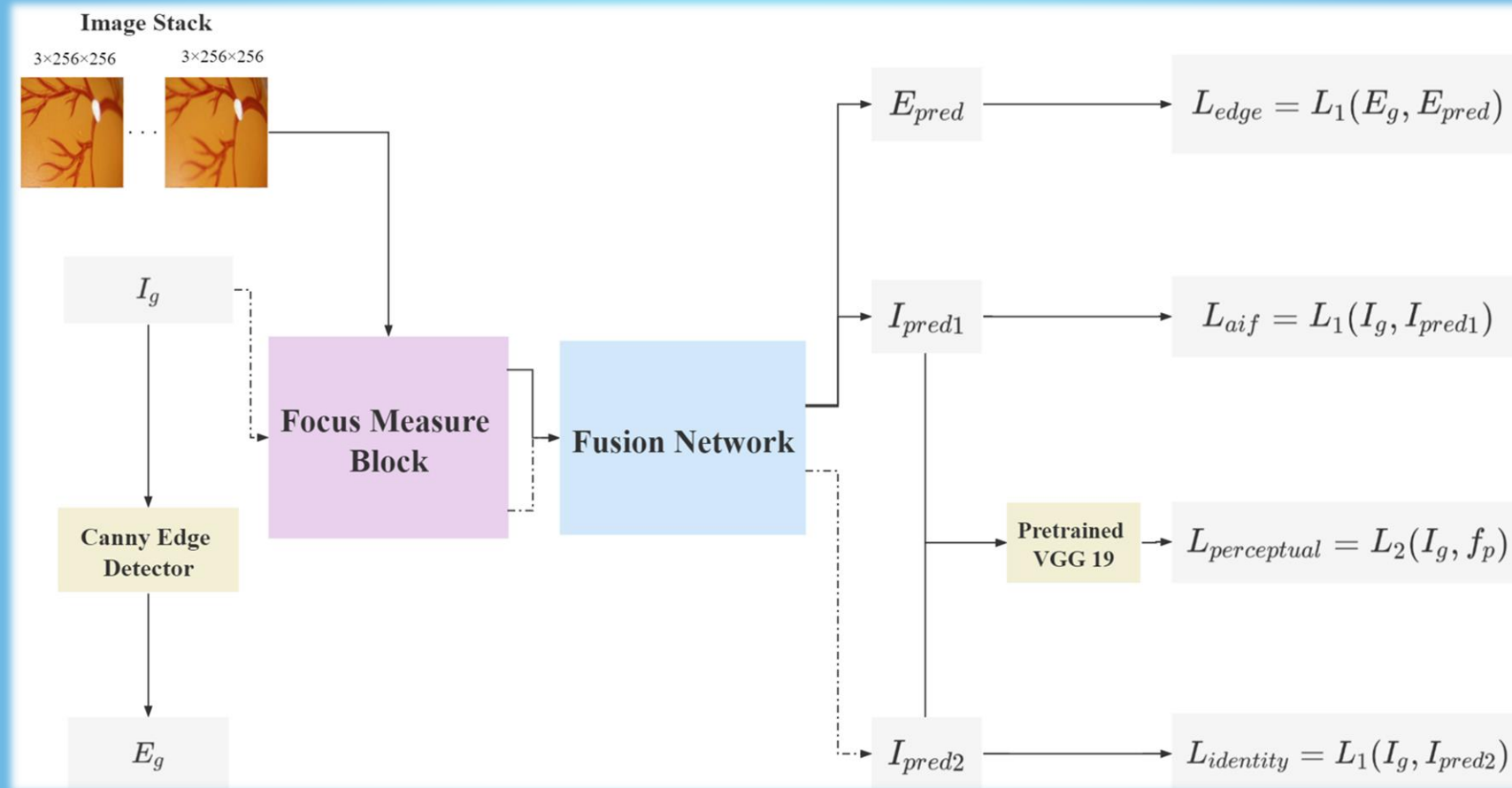
End-to-end CNN Network

Focus Measure Block extracts the focus degrees of each pixel

Edge Loss and Identity Loss regularize the network to maintain contextual details and color mapping

Methodology (Multi-focus Image Fusion)

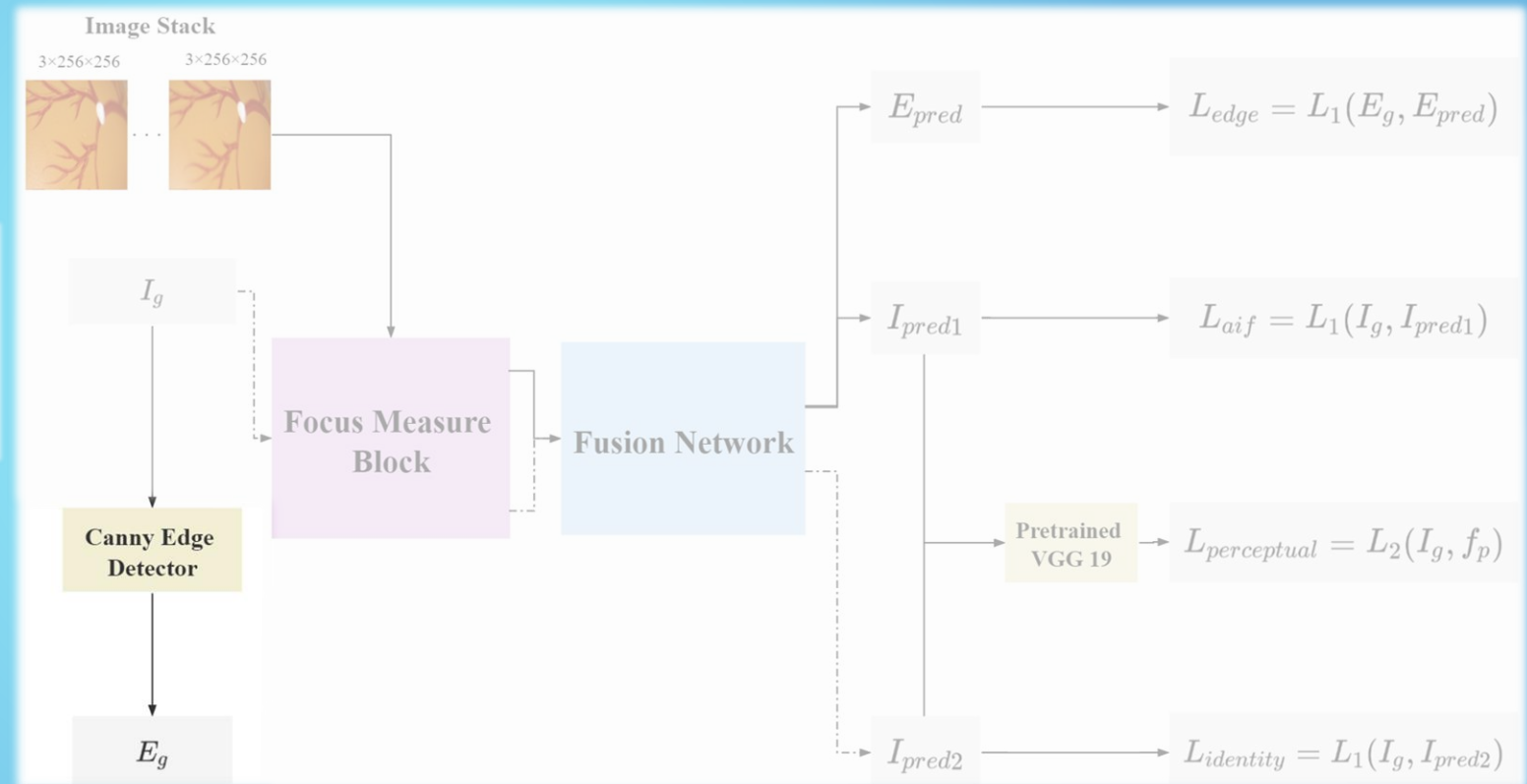
Focal Stack with
different out-of-
focused areas



One all-in-
focus image

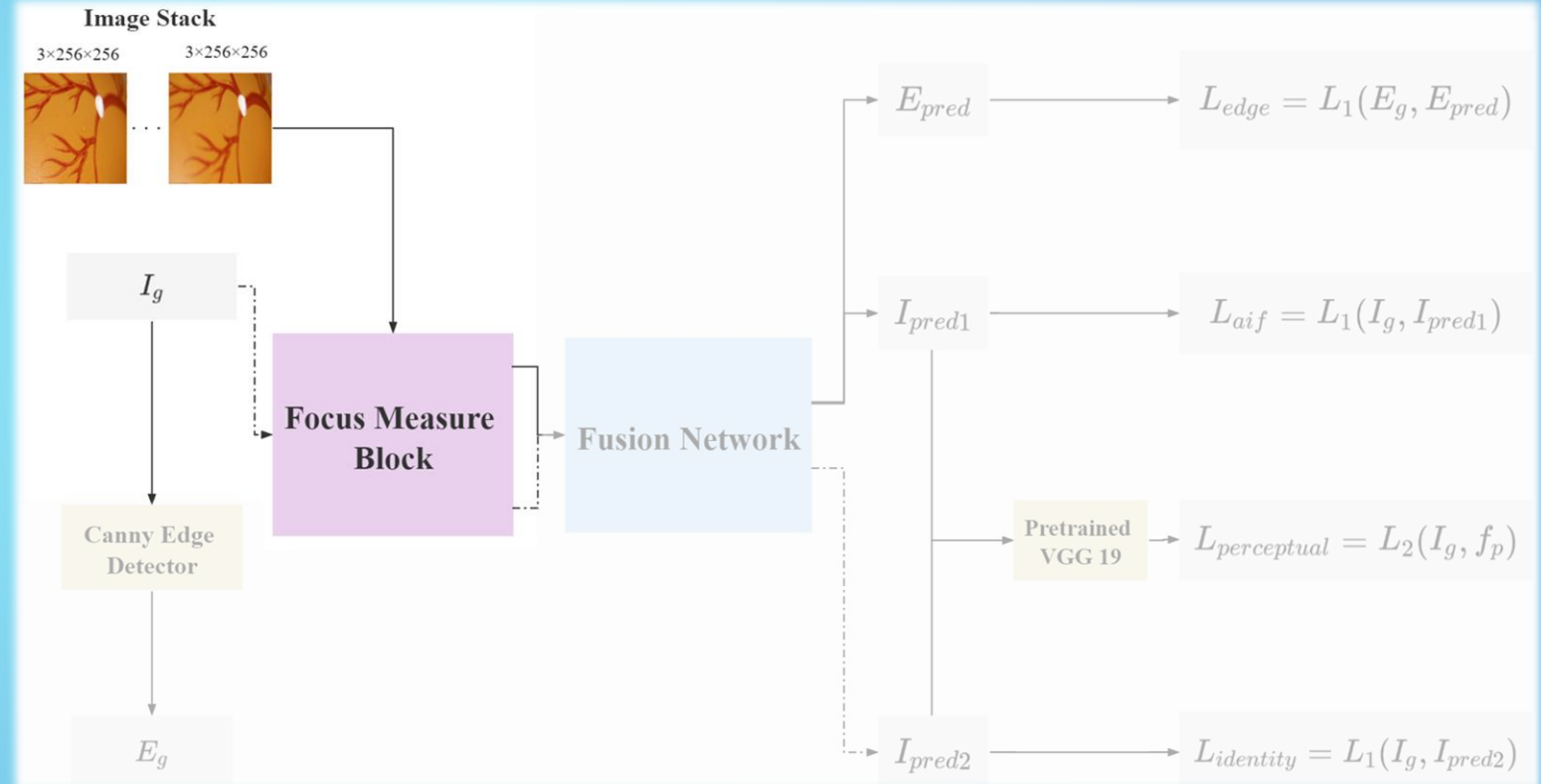
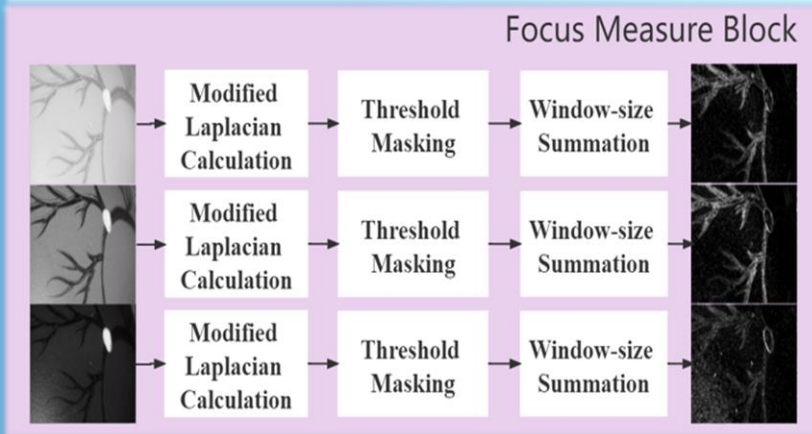
Methodology (Edge Detector)

1. Detects **edge information** from the ground-truth images
2. Saves for the calculation of edge loss



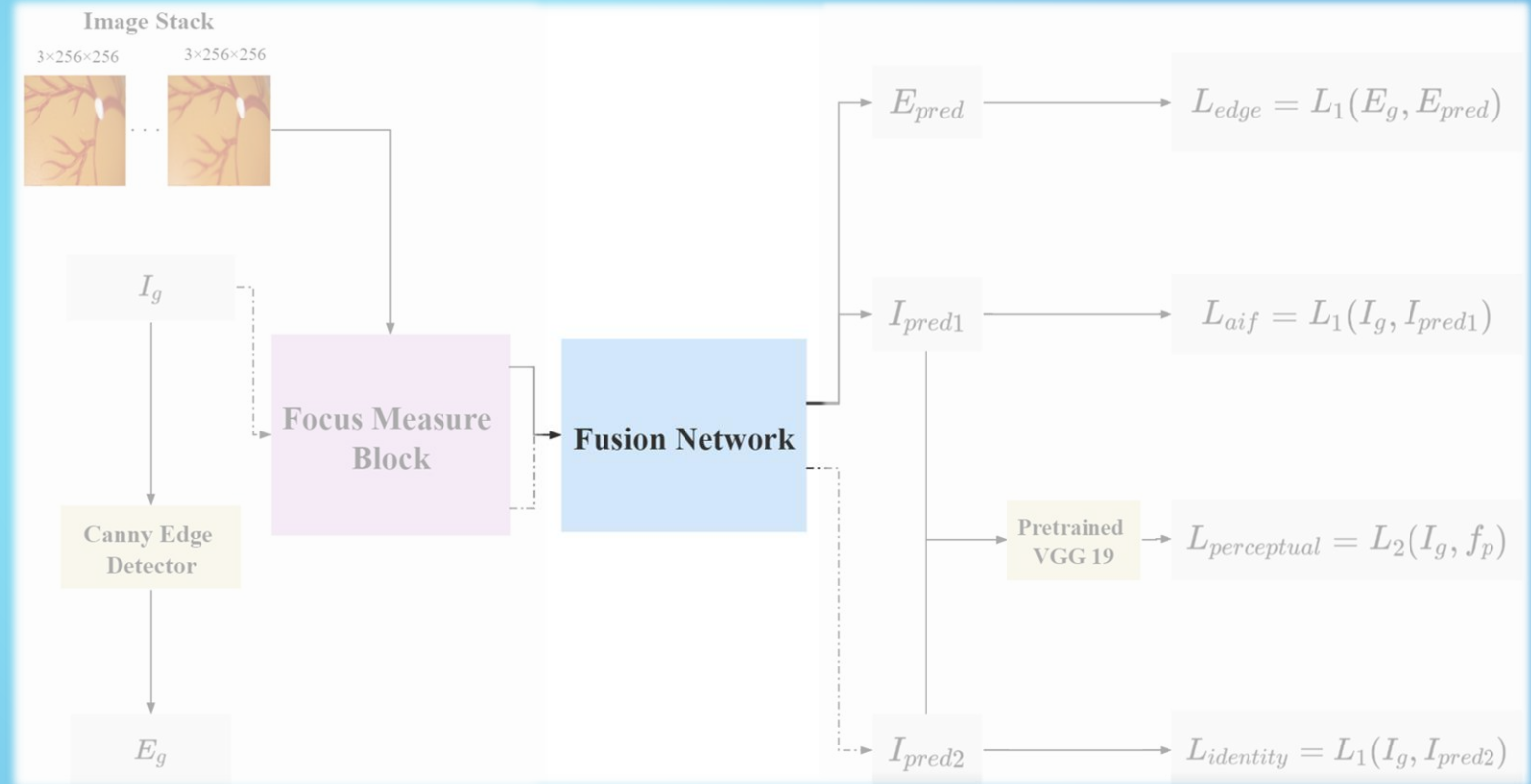
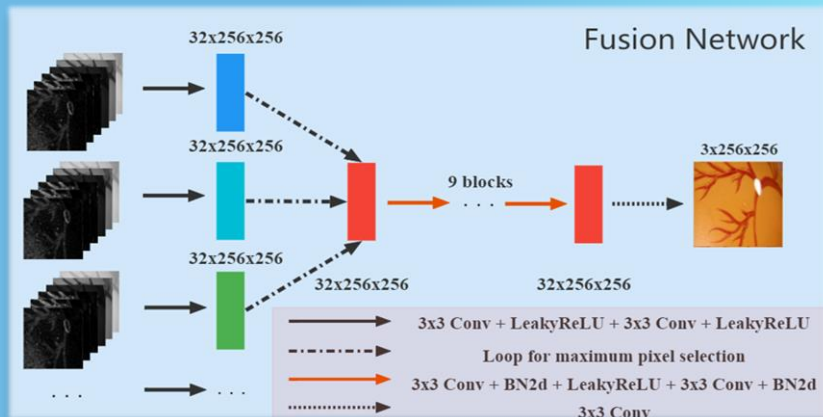
Methodology (FMB)

1. Implements sum-modified-laplacian
2. Extracts channel-by-channel **focus measure**
3. Concatenates input images with their focus measure



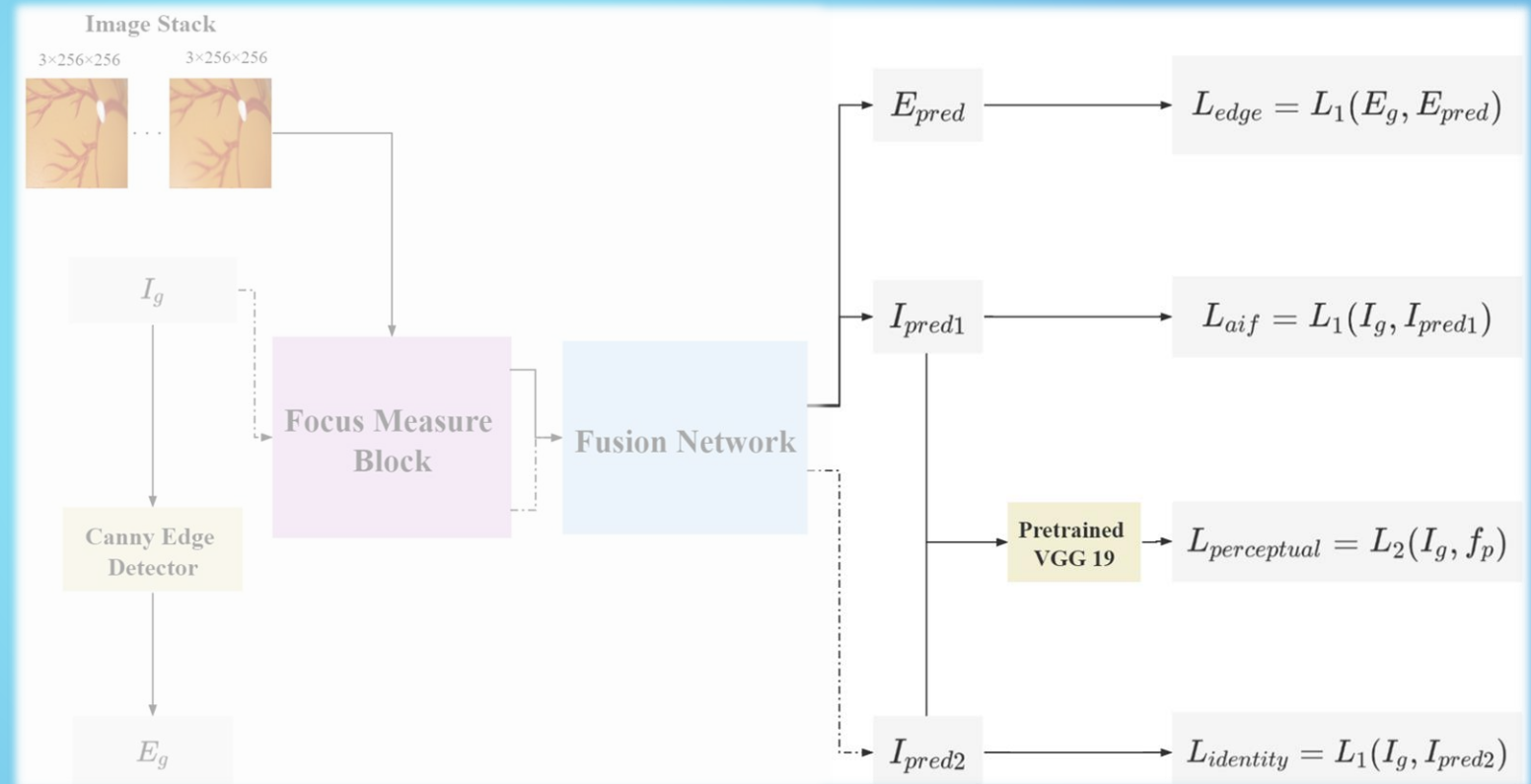
Methodology (FN)

1. Samples features from the defocused image stacks
2. Fuses feature maps and iteratively selects maximum pixel values
3. Reconstructs the all-in-focus images



Methodology (FN)

1. Aif Loss and Perceptual Loss: to avoid the smoothing effects of MSE Loss
2. Edge Loss: to maintain **edge's sharpness**
3. Identity Loss: to maintain the same color mapping between inputs and outputs



Experiments and Results

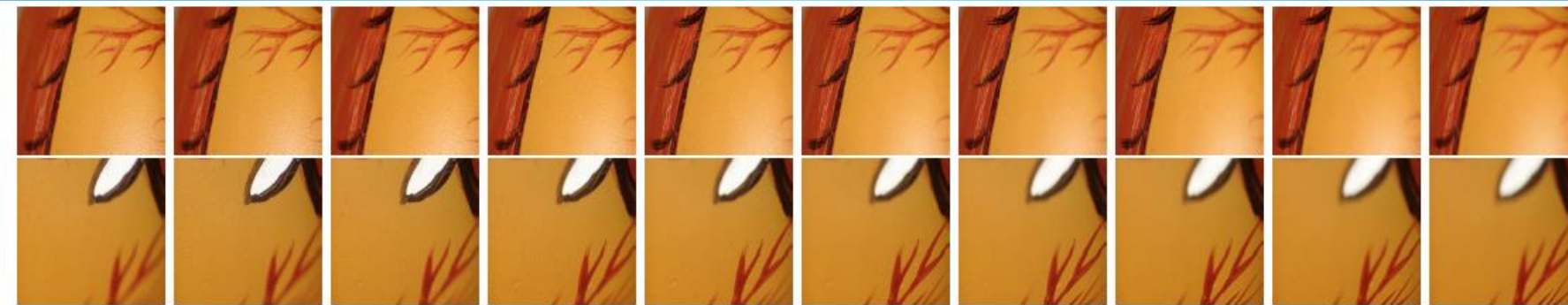
Ablation Study for the effectiveness of FMB and the loss

Model	FN	FN+FMB ₁	FN+FMB ₃	FN+FMB ₃ + Edge loss	FN+FMB ₃ + Identity loss	Ours
SSIM	0.8942	0.8860	0.9071	0.9079	0.9193	0.9230
PSNR	30.64	29.86	30.80	31.30	30.82	32.10

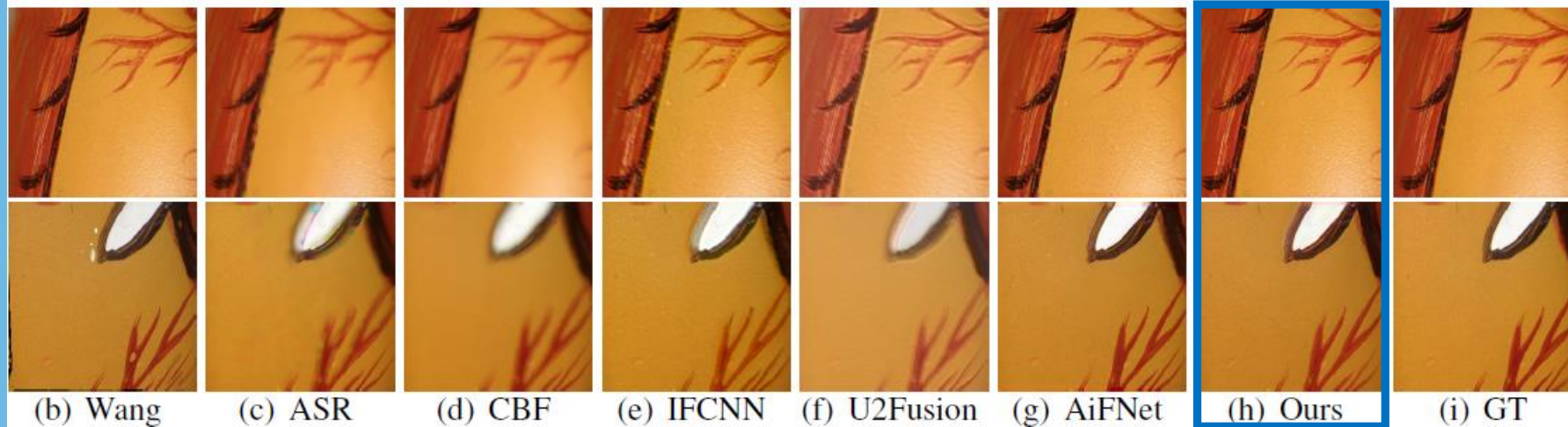
Comparative Experiments with other models

Model	Wang	ASR	CBF	IFCNN	U2Fusion	AiFNet	Ours
SSIM	0.6782	0.8054	0.7180	0.8583	0.7620	0.9054	0.9230
PSNR	25.04	29.26	26.43	29.35	25.64	31.91	32.10

Experiments and Results



(a) Image stack with different defocused area.



(b) Wang

(c) ASR

(d) CBF

(e) IFCNN

(f) U2Fusion

(g) AiFNet

(h) Ours

(i) GT

Application (Panoramic View of the Fundus)



Better panoramic images (left) from generated all-in-focus images and image stitching algorithms than using defocused image stacks (right)

Conclusion

A novel multi-focus image fusion model

Focus Measure Block that obtains the focused areas

Fusion Network that fuses defocused images

Edge Loss that maintains the boundary, edge, and contextual information

Experiments that proves our models' effectiveness

Future Work

To improve our model's performance on **real-world** optic microsurgery images and videos



Thank you for listening!

